

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (Civil /Mechanical) (CBCS) Degree Examinations

December 2023

MA2003-1 – VECTOR CALCULUS & COMPLEX FUNCTIONS

(For CV, ME)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. If $\vec{F} = 3\hat{i} + 3y\hat{j} + 3z\hat{k}$ then $\nabla \cdot \vec{F} =$ _____
A) 3 B) 0
C) 6 D) 1
2. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\nabla \times \vec{r} =$ _____
A) 0 B) 2
C) 1 D) 3
3. If $\phi = x + y + z$, then $\nabla \phi =$ _____
A) $\hat{i} + \hat{j} + \hat{k}$ B) 0
C) $4\hat{j}$ D) 1
4. If $\vec{F} = 7z\hat{i} + 5x\hat{j} + 2y\hat{k}$ then the value of $(\nabla \times \vec{F}) =$ _____
A) 0 B) $2\hat{i} - 7\hat{j} - 5\hat{k}$
C) $2\hat{i} - 7\hat{j} + 5\hat{k}$ D) $2\hat{i} + 7\hat{j} + 5\hat{k}$
5. By the Stoke's theorem, relation between the line integral and the surface integral is _____
A) $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \vec{F} \cdot \hat{n} dS$ B) $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \nabla \cdot \hat{n} dS$
C) $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \phi) \cdot \hat{n} dS$ D) $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$
6. By the Gauss divergence theorem, relation between the surface and the volume integral is _____
A) $\oint_C \vec{F} \cdot d\vec{r} = \iint_S \nabla \cdot \hat{n} dS$ B) $\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V \nabla \times \vec{F} dV$
C) $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$ D) $\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V \nabla \cdot \vec{F} dV$
7. The function $f(z) = \log(z)$ is _____
A) Analytic everywhere B) Analytic no where
C) Differentiable everywhere D) differentiable everywhere except at the origin
8. Stoke's theorem uses which of the following operations?
A) Gradient B) Divergence
C) Curl D) Laplacian
9. A bilinear transformation $w = \frac{az+b}{cz+d}$ maps circles to _____
A) squares B) circles
C) triangles D) Straight lines

10. A function $f(z)$ satisfying the Cauchy-Riemann equations is known as _____ function.
 A) Analytic B) Non - Analytic
 C) Harmonic D) Non - Harmonic
11. Cauchy-Riemann equations in polar form are _____
 A) $u_r = \frac{v_\theta}{r}, \frac{u_\theta}{r} = v_r$ B) $u_r = \frac{v_\theta}{r}, \frac{u_\theta}{r} = -v_r$
 C) $u_r = -\frac{v_\theta}{r}, \frac{u_\theta}{r} = -v_r$ D) $u_r = v_r, u_\theta = -v_\theta$
12. A function $u = 3x + 4y$ satisfying the Laplace equation is _____
 A) Non-analytic B) Harmonic
 C) Non-harmonic D) Analytic
13. For the function $f(z) = \frac{z^2+5}{z^2(8z+5)}$ the pole $z = 0$ has order _____
 A) 1 B) 3
 C) 2 D) 4
14. A circle centered at a point $z = 2$ and having radius 5 is denoted by
 A) $|z| = 5$ B) $|z + 2| = 5$
 C) $z - 5 = 2$ D) $|z - 2| = 5$
15. If $f(z) = \frac{e^{2z}}{z^2-1}$ then the poles of z are _____
 A) ± 1 B) $\pm i$
 C) 1, 4 D) 3, 1
16. If $f(z) = \frac{2z-7}{(z+2)(z-3)}$ then $z = 3$ is known as _____
 A) Pole of $f(z)$ B) Residue of $f'(z)$
 C) Zero of $f(z)$ D) Residue of $f(z)$
17. Which of the following functions is even?
 A) $x^2 + x^4$ B) $\sin 5x$
 C) $x^3 + x$ D) $\tan x$
18. If $f(x)$ is a periodic function in $(-\pi, \pi)$, then $a_n =$ _____
 A) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$ B) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$
 C) $\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$ D) $\int_{-\pi}^{\pi} f(x) \cos nx dx$
19. Fourier transform of the function $g(x) =$ _____
 A) $\int_{-\infty}^{\infty} g(x) e^x dx$ B) $\int_{-\infty}^{\infty} g(x) e^{-sx} dx$
 C) $\int_{-\infty}^{\infty} g(x) e^{-isx} dx$ D) $\int_{-\infty}^{\infty} g(x) e^{sx} dx$
20. Which of the following functions is odd?
 A) $f(x) = \begin{cases} 1+x, & -\pi < x < 0 \\ 1-x, & 0 < x < \pi \end{cases}$ B) $f(x) = \begin{cases} -x, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$
 C) $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$ D) $f(x) = \begin{cases} -2x, & -\pi < x < 0 \\ 2x, & 0 < x < \pi \end{cases}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1. a)	Find $\nabla \phi$ and $ \nabla \phi $ for $\phi = 2x^3y^2z^4$ at $(3, -1, 2)$.	4	L*1	1	1
b)	Prove that the spherical co-ordinate system is an orthogonal curvilinear coordinate system.	6	L2	1	1

- c) Using Gauss divergence theorem evaluate: $\iint_S \vec{F} \cdot \hat{n} dS$, where $\vec{F} = (y - z + 2)\hat{i} - (yz + 4)\hat{j} - (zx)\hat{k}$, and S is the closed surface of the cube $x = 0, x = 2, y = 0, y = 2, z = 0, z = 2$. 6 L2 2 1
2. a) A particle moves along a curve $x = 7 + 8t^5, y = 4t^2$ and $z = 5 + t$ where t is the time variable. Find the components of velocity and acceleration at $t=2$ in the direction $\hat{i} + 3\hat{j} + \hat{k}$. 4 L1 1 1
- b) State and prove Greens theorem in a plane. 6 L2 1 1
- c) If $\phi = 2x^3y^2z^4$ and $\vec{F} = \nabla\phi$, find $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$. 6 L2 2 1
3. a) Determine the constants a, b, c so that the vector function $\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational. 4 L1 1 1
- b) Find the work done in moving a particle once round the circle C in the xy-plane with centre at the origin and radius 3, in the force field: $\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y)\hat{k}$. 6 L2 2 1
- c) Using Stoke's theorem evaluate: $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (x^2 + y^2)\hat{i} - (2xy)\hat{j}$ round the rectangle C in the plane $z = 0$ and bounded by the lines $x = 0, y = 0, x = a$ and $y = b$. 6 L2 2 1

Unit – II

4. a) Define a harmonic function, check whether the function: $u = x^3 - 3xy^2 + 3x^2 - 3y^2$ is harmonic. 4 L2 3 1
- b) If $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D, then prove that at any point of D, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. 6 L2 3 1
- c) State Cauchy's Residue theorem and use it to evaluate: $\int_C \frac{(z^2)dz}{(z-1)^2(z+2)}$ where C is the circle $|z| = \frac{5}{2}$. 6 L4 4 1
5. a) Show that $f(z) = \cos(z)$ is an analytic function and find $f'(z)$. 4 L3 3 1
- b) Find the poles and residues of $f(z) = \frac{z^2}{(z^2+3z+2)^2}$. 6 L2 4 1
- c) Write the Laurent's series which represent: $f(z) = \frac{z+3}{z(z+1)(z-2)}$ in i) $|z| < 1$, ii) $1 < |z| < 2$. 6 L2 4 1
6. a) Find the bilinear transformation which maps $z = 1, i, -1$ to $w = 2, i, -2$. 4 L3 3 1
- b) Using Cauchy's integral formula evaluate $\int_C \frac{zdz}{(z-5)(2z+3)}$ where C is the circle $|z| = 4$. 6 L2 4 1
- c) State and prove Cauchy's Integral theorem. 6 L2 3 1

Unit – III

7. a) Obtain the half range Fourier sine series of $f(t) = t - t^2$ in the interval $0 < t < 1$. 4 L3 5 1
- b) Obtain the first two coefficients in the Fourier cosine series of y from the following table.
- | | | | | | | |
|---|---|---|----|---|---|---|
| x | 0 | 1 | 2 | 3 | 4 | 5 |
| y | 4 | 8 | 15 | 7 | 6 | 2 |
- c) Obtain the Fourier series of $f(x) = \begin{cases} 0 & -2 < t < -1 \\ k & -1 < t < 1 \\ 0 & 1 < t < 2 \end{cases}, f(t+4) = f(t)$. 6 L2 5 1
8. a) Find the Fourier sine transform of $f(x) = e^{-|x|}$. 4 L1 5 1
- b) Find the Fourier transform of $f(x) = e^{-a^2x^2}, a > 0$. 6 L2 5 1
- c) State and prove convolution theorem of Fourier transforms 6 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
